

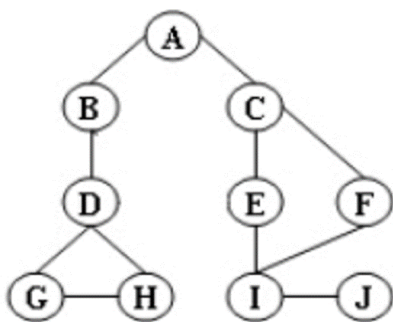
2020-2021 学年第一学期期末考试试卷

一、判断题(每小题 2 分, 共 20 分)

1. () The average run time and the extra space of Heapsort for sorting n elements are $O(n \log n)$ and $O(1)$ respectively.
2. () Consider two programs with time complexities being $T1 = O(2^n)$ and $T2 = O(n \log n)$, then program 2 must run faster than program 1.
3. () The inorder traversal sequence of any max-heap must be in sorted order.
4. () In most restaurants, we follow one principle called "First come, first served". This principle can be implemented by a queue.
5. () If a general tree T is converted into a binary tree BT , then the BT 's post-order traversal has the same sequence as that of the pre-order traversal of T .
6. () For a binary tree, if its pre-order travel sequence is $\{4, 2, 1, 3, 6, 5, 7\}$, and its in-order travel sequence is $\{1, 2, 3, 4, 5, 6, 7\}$, then 4 is the parent of 3.
7. () For a connected graph, if there are exactly two vertices having odd degree, we can find an Euler circuit that visits every vertex exactly once by starting from one of its odd-degree vertices.
8. () During sorting $(25, 21, 47, 15, 27, 68, 35, 20)$ in ascending order (升序) with the iterative version of Mergesort, $(15, 21, 25, 47, 20, 27, 35, 68)$ is the result of the second run.
9. () The storage size of a graph using the adjacency list is only related to the number of vertices but has nothing to do with the number of edges.
10. () In hashing with open addressing to solve collisions, the operation FIND will be seriously slowed down if there are too many deletions intermixed with insertions.

二、选择题(每小题 3 分, 共 60 分)

- 1、The articulation points of the given graph are: ()



- M. A, B, C, D
 N. A, B, C, D, I
 O. F, E, D
 P. A, B, I

2、 Given the adjacency list of a directed graph.

0: 4, 5, 6

1:Empty

2:Empty

3:1, 2

4:Empty

5:3

6:2

Which one below is NOT a valid topological order of the graph?()

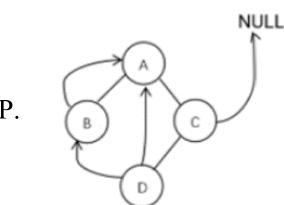
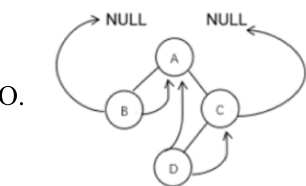
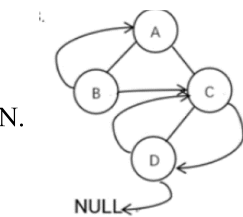
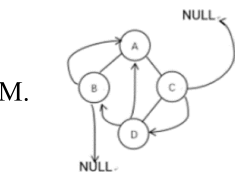
E. 0 5 4 3 6 1 2

F. 0 4 5 3 1 6 2

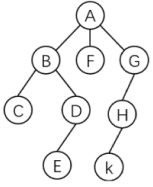
G. 0 6 5 4 3 2 1

H. 0 4 6 2 5 3 1

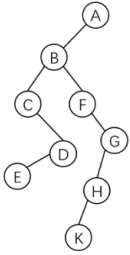
3、 Among the following threaded binary trees (the threads are represented by arrows), which one is the pre-order threaded tree?()



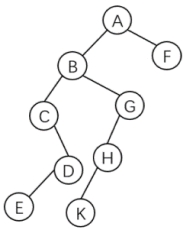
4、The following figure shows a tree. Which one is its corresponding binary tree with the "first—child/next sibling" representation?()



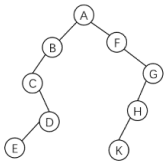
M.



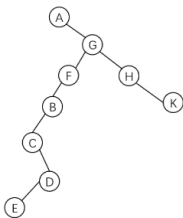
N.



O.



P.



5、Let's traverse a complete binary tree in level—order, and define the level—order index of the root to be 1. Then for the i-th node visited, if its left child exists, then the index of this left child is ()

M. $i+1$

N. $2i-2$

O. $2(i+1)+2$

P. $2i$

6、 It is known that the fifth layer of a complete binary tree (the root is the first layer) has 8 leaf nodes. then the number of nodes of the complete binary tree is at most ()

E. 39

F. 31

G. 52

H. 47

7、 In a weighted undirected graph, if the length of the shortest path from v_1 to v_0 is 14, and there exists an edge of weight 1 between v_2 and v_1 , then which one of the following is correct?()

M. The length of the shortest path from v_2 to v_0 must be less than 15

N. The length of the shortest path from v_2 to v_0 must be no less than 13

O. The length of the shortest path from v_2 to v_0 must be greater than 15

P. The length of the shortest path from v_2 to v_0 must be no greater than 13

8、 Given a binary tree, if its Pre—order traversal is $\{A,B,D,E,F,G,C\}$ and the post-order traversal is $\{D,F,G,E,B,C,A\}$, then which one is its In-order traversal?()

E. BDFGECA

F. DBFEGAC

G. BDEFGAC

H. DFGEBCA

9、 An inversion in an array $A[]$ is any ordered pair (i,j) having the property that $i < j$ but $A[i] > A[j]$. Given array $A: \{3, 87, 12, 61, 70, 26, 45\}$, after the first partition of Quicksort with Median3pivot selection, the number of inversions will be decreased by()

M. 6

N. 9

O. 3

P. 1

10、 Given a hash table of size 13 (indexed from 0 to 12) with the hash function $H(\text{Key}) = \text{Key} \% 11$, Quadratic probing $H(\text{Key}) = (H(\text{Key}) + i * i) \% 13$ is used to resolve collisions when the i -th ($i > 0$) collision occurs. Then after inserting $\{10, 21, 32, 33, 65, 12\}$ one by one into the hash table, which one of the following statements is false? ()

E. the key 12 is at position 1

F. the loading density is less than 0.5

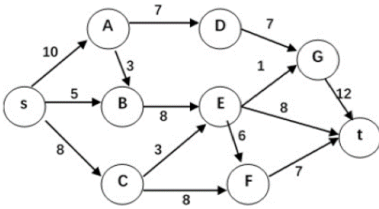
G. the key 65 is at position 6

H. the average search time is greater than 2

11、 In hashing with open addressing method, rehashing is definitely necessary when ()

- E. primary clustering occurs
- F. the hash table is half full
- G. the hash function is not uniform
- H. an insertion fails

12、The maximum flow in the network of the given Figure is: ()



- M. 20
- N. 21
- O. 22
- P. 23

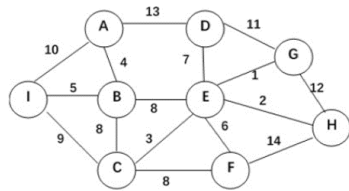
13、Suppose A is an array of length N with some random numbers. What is the time complexity of the following program in the worst case? ()

```

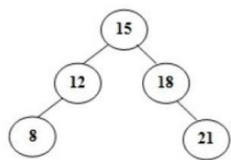
void function( int A[], int N ) {
    int i, j = 0, cnt = 0;
    for (i = 0; i < N; ++i) {
        for (; j < N && A[j] <= A[i]; ++j);
        cnt += j - i;
    }
}
  
```

- A. $O(N \log N)$
- B. $O(N^2)$
- C. $O(N^{1.5})$
- D. $O(N)$

14、 Given a weighted graph as shown by the figure. Which one of the following statements is TRUE about its minimum spanning tree? ()



- A. If the weight of edge (B,C) was 7, then the minimum spanning tree would be unique.
 - B. If the weight of edge (B,E) was 7 then the minimum spanning tree would not be unique
 - C. The minimum spanning tree is not unique and the total weight is 35.
 - D. The minimum spanning tree is unique and the total weight is 36.
- 15、 Which one of the following is impossible to be the insertion sequence of the given binary search tree? ()

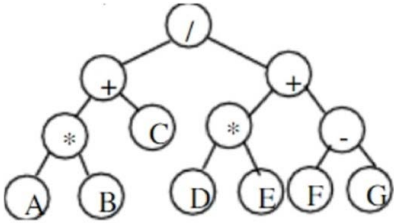


- A. 15, 18 ,21, 12, 8
 - B. 15, 18, 12, 21,8
 - C. 15, 12 ,8 ,18, 21
 - D. 15, 12, 21, 8, 18
- 16、 Given $S = \{ 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 \}$ and 9 equivalence relations: 3~4, 6~8, 9~11, 2~3, 1~11, 3~11, 5~8, 1~7, 11~4. After invoking successively these relations with union-by—size (if the two sizes are equal. the smaller element will be the root) and path compression, which one of the following statements is false? ()
- A. there are 3 equivalence classes
 - B. all elements are either a root or a child of a root
 - C. the parent of element 1 is element 3
 - D. the parent of element 11 is element 9
- 17、 One of the following algorithms: Selection sort, Shell sort. Insertion sort. is applied to sort the sequence (2, 12, 16, 88, 1,5,10) in ascending order. if the resulting sequences of the first two runs are (2, 12,16, 10, 1, 5, 88) and (2, 12, 5, 10, 1, 16, 88). then after the third run, the number in front of 5 must be ()
- A. 1
 - B. 12

C. None

D. 2

18、The following binary tree is called an expression tree. Which one is the arithmetic expression that this tree represents? ()

A. $(A*B+C)/(D*E+(F-G))$ B. $A*B+C/(D*E)+(F-G)$ C. $A*B+C/D*E+F-G$ D. $(A*B+C)/(D*E)+(F-G)$

19、The following list IS a series of operations for a stack

- push 1

- push 2

- pop

- push 3

- pop

- push 4

- push 5

- pop

- pop

- pop

- push 6

- push 7

- pop

- pop

- push 8

- pop

- push 9

- pop

Which is the correct pop-up sequence? ()

- A. 2 3 5 4 7 6 8 9 1
- B. 1 2 3 4 5 6 7 8 9
- C. 9 8 7 6 5 4 3 2 1
- D. 2 3 5 4 1 7 6 8 9

20、 Given an array of integers { 18, 28, 32, 22, 8, 10, 30 }. Build a min—heap using the linear algorithm and then call DeleteMin twice. Which of the following is the level—order traversal sequence of the remaining heap? ()

- A. 18 22 30 32 28
- B. 18 30 28 22 32
- C. 18 22 30 28 32
- D. 18 30 22 32 28

三、程序填空题（满分 12 分）

1、 (11 分) The function : CountShortestPaths is to find the number of shortest paths from Vertex S to every other vertices in a given graph (positive weights only). The distances and the numbers of shortest paths are stored in dist[] and count[], respectively. The Graph is defined as follows:

```
typedef struct GNode *PtrToGNode;
struct GNode{
    Int Nv; /* Number of vertices */
    Int Ne; /* Number of edges */
    Int AdjMat[MaxVertexNum][MaxVertexNum]; /* adjacency matrix */
}.
typedef PtrToGNode Graph;
```

please fill in the blanks.

```
void CountShortestPaths(Graph G, Vertex S, bool known[], int dist[], int count[])
{
    for (int i = 0; i < G->Nv; ++i)
    {
        Known[i]=false;
        dist[i]=INFINITY;
        count[i]=0;
    }
    dist[S]=0;
    _____;(3 分)
    while(true){
        Vertex V = FindSmallestUnknown(G, known, dist);
        if(V== -1)break;
```



```

known[V]=true;
for(Vertex W=0;W<G->Nv;++W){
    int weight = G->AdjMat[V][W];
    if(weight && !known[W]){
        if(dist[V]+weight < dist[W]){
            dist[W]=dist[V]+weight;
            count[W]=count[V];
        } else if (dist[V]+weight == dist[W]){
            _____;(3 分)
        }
    }
}
}
}
}
}

```

2、(2 分) Following function Shellsort(int A[], int N) is the implementation of Shellsort Algorithm with increment sequence (1,3,7,11). **Please fill in the blanks.**

```

void Shellsort(int A [ ], int N)
{
    int i, j, k, Increment, Inc[ ]={1,3,7,11};
    int Tmp;
    for(k=3;k>=0;k--)
    {
        _____;(3 分)
        for (i=Increment;i<N;i++)
        {
            Tmp=A[i];
            for(j=i;j>=Increment;j-=Increment)
                if(Tmp<A[ j-Increment])
                    _____;(3 分)
            Else break;
            A[j]=Tmp;
        }
    }
}

```

四、函数题（8分）

You are supposed to write a function of finding the height of a binary search tree with the given preorder sequence.

int Height_of_BST(int preorder[],int N);

where the preorder sequence is stored in int preorder[], and the integer N is the number of nodes in the tree, which is guaranteed to be positive. The function Height_of_BST is supposed to return the height of the binary search tree.

Note:

- It is guaranteed that the preorder sequence consists of distinct integers and does correspond to a binary search tree.
- You may assume that MAXN is a small number (less than 100) in the Judge's program.

Sample program of judge:

```
#include<stdio.h>
#include<stdlib.h>
#define MAXN 10
int Height_of_BST(int preorder[],int N);
int main()
{
    int preorder[MAXN],N,i;
    scanf("%d",&N);
    for(i=0;i<N;i++)scanf("%d",&preorder[i]);
    printf("%d\n",Height_of_BST(preorder,N));
    return 0;
}
/*Your function will be put here*/
```

Sample Input:

```
9
10 6 2 9 8 25 20 22 30
```

Sample Output:

```
3
```

2020-2021 学年第一学期期末考试试卷参考答案

一、判断题(每小题 2 分, 共 20 分)

1、【正解】T

【解析】在堆排序中先进行一次堆的初始化操作, 其时间复杂度为 $O(n)$, 每次调整的时间复杂度为 $O(\log n)$, 排序共需 $n-1$ 次调整。所以, 堆排序的时间复杂度为 $O(n \log n)$ 。堆排序可以不用多余的存储空间, 因此空间复杂度为 $O(1)$

【考点延伸】堆排序

2、【正解】F

【解析】大 O 时间复杂度实际上并不具体表示代码真正的执行时间, 而是表示代码执行时间随数据规模增长的变化趋势。

【考点延伸】时间复杂度

3、【正解】F

【解析】最大堆只保证父结点的值大于其左右孩子的结点, 而左右孩子之间并没有大小关系。故采用中序遍历最大堆时, 输出的序列不一定有序

【考点延伸】最大堆, 二叉树的遍历

4、【正解】T

【解析】队列是一种特殊的线性表, 特殊之处在于它只允许在表的前端 (front) 进行删除操作, 而在表的后端 (rear) 进行插入操作, 即先进先出。

【考点延伸】队列

5、【正解】F

【解析】二叉树前序遍历先访问当前结点, 再访问其左右孩子结点, 后序遍历则是先访问其左右孩子结点, 再访问当前结点, 二者的遍历序列一定不同。

【考点延伸】二叉树的遍历

6、【正解】F

【解析】由前序遍历和后序遍历可确定唯一的二叉树, 4 是二叉树的根结点, 而 3 是叶子结点。

【考点延伸】二叉树的遍历

7、【正解】F

【解析】如果一个连通图的所有顶点的度数都为偶数, 那么这个连通图具有欧拉回路, 且这个图被称为欧拉图。欧拉路径是图中的一条路径, 该路径满足 恰好访问每个边一次。而欧拉回路是一条在同一顶点处开始和结束的欧拉路径。

【考点延伸】欧拉路径和欧拉图

8、【正解】T

【解析】迭代版本的归并排序先分组再合并排序, 第一趟是每两个内部排序, 排序结果是 21 25 15 47 27 68 20 35, 第二趟是每四个内部排序, 结果是 15 21 25 47 20 27 35 68。

【考点延伸】归并排序

9、【正解】F

【解析】邻接表是结构体数组加链表的结合体，数组存储顶点，每一个顶点的链表存储与该顶点相连的所有顶点，边越多，需要开辟的链表越长，所需存储空间越大。

【考点延伸】邻接表

10、【正解】T

【解析】用开放寻址法解决冲突的散列表，删除数据的时候比较麻烦，需要特殊标记已经删除掉的数据。而且，在开放寻址法中，所有的数据都存储在一个数组中，比起链表法来说，冲突的代价更高，会使 FIND 操作严重减慢

【考点延伸】散列冲突

二、单选题(每小题 3 分，共 60 分)

1 【正解】B

【解析】删除 ABCDI 中任一个结点都会使该图不连通

【考点延伸】连通图

2 【正解】D

【解析】结点 3 是指向结点 2 的，因此 2 不能在 3 之前输出

【考点延伸】拓扑排序

3 【正解】B

【解析】前序：ABCD，结点的左指针指向前序中前一个元素，右指针指向前序中后一个元素

【考点延伸】二叉树的遍历

4 【正解】A

【解析】任一结点的左孩子是其孩子结点，右孩子是其兄弟结点。故 A 正确

【考点延伸】树的表示

5 【正解】D

【解析】将完全二叉树放入数组中，根结点为 1，则对于任一个结点，如果左孩子存在，则其下标为 $2*i$

【考点延伸】完全二叉树

6 【正解】D

【解析】最多有 $1+2+4+8+16+16=47$ 个结点

【考点延伸】完全二叉树

7 【正解】B

【解析】 $V_1V_0 \leq V_1V_2 + V_2V_0$, 因此 V_2 到 V_0 的最短路一定大于等于 13

【考点延伸】最短路径

8 【正解】B

【解析】中序遍历为 DBFEGAC

【考点延伸】二叉树的遍历

9 【正解】A

【解析】3 87 12 61 70 26 45 逆序对为 9 在第一次分区后为 3 26 12 45 70 87 61
逆序对为 3，故减少 6

【考点延伸】快速排序

10 【正解】B

【解析】33 在 0，12 在 1，32 在 2，65 在 5，10 在 10，21 在 11. 装载因子小于 0.5。平均搜索时间大于 2

【考点延伸】散列表

11 【正解】D

【解析】当插入失败时说明哈希表已满，需要扩容并重新计算哈希值。

【考点延伸】散列冲突

12 【正解】D

【解析】从 s 出发的流量 10 5 8, 都能到达 t, 故最大流量为 23

【考点延伸】最大流

13 【正解】D

【解析】注意 j 在循环开始时没有赋初值, 内部的 for 循环在整个函数中最多只会被执行 N 次, 外循环会执行 N 次, 因此时间复杂度为 $O(n)$

【考点延伸】时间复杂度

14 【正解】A

【解析】A. 当前的最小生成树不唯一, 因为 BC 和 BE 的权值相同, 两者任一皆可选择。如果 (B,C)是 7, 则 BC 一定相连, 最小生成树唯一 C. $1+2+3+4+5+6+7+8=36$

【考点延伸】最小生成树

15 【正解】D

【解析】D 选项中 21 会成为 15 的右孩子, 与题不符

【考点延伸】二叉搜索树

16 【正解】D

【解析】经过插入这些关系和路径压缩可以得到三个集合 $\{10\}, \{6, 8, 5\}, \{3, 1, 9, 7, 2, 4, 11\}$, 第一个均为根, 其余结点都直接连接在根结点上, 因此 11 的父结点是 3

【考点延伸】并查集, 路径压缩

17 【正解】A

【解析】从两次排序结果可以看出该排序算法是简单选择排序, 下一轮从 2 12 5 10 中选择一个最大的 12 与 1 交换得到 2 1 5 10 12 16 88, 因此 5 之前是 1

【考点延伸】简单选择排序, 插入排序, 希尔排序

18 【正解】A

【解析】中序遍历该二叉树, 左右子树的运算加上括号即可

【考点延伸】二叉树的遍历

19 【正解】D

【解析】栈是先进先出的数据结构, 模拟出入栈即可

【考点延伸】栈

20 【正解】A

【解析】用数组存储的堆是一个完全二叉树, 将父结点和左右孩子分别比较, 与三者中最小的交换, 从底部依次往上调整, 得到最小堆。删除操作是从上往下调整, 调整算法类似。

【考点延伸】最小堆

三、程序填空题 (满分 12 分)

1、【解析】(1) `count[S]=1;` //到自身有一条路

(2) `count[W]+=count[V]` //从 V 到 W 有 `count[V]` 条路, 还需加上本身的 `count[W]`

【考点延伸】dijkstra 算法, 单源最短路

2、【解析】(1) `Increment=Inc[k]` //插入排序的增量

(2) `A[j]=A[j-Increment]` //插入排序

【考点延伸】希尔排序

四、函数题(满分 8 分)

【解析】

```
#include<stdio.h>
```

```
#include<stdlib.h>
```

```
#define MAXN 10
```

```
int Height_of_BST(int preorder[],int N);
```

```
int main()
```

```
{
    int preorder[MAXN],N,i;
    scanf("%d",&N);
    for(i=0;i<N;i++)scanf("%d",&preorder[i]);
    printf("%d\n",Height_of_BST(preorder,N));
    return 0;
}
/*Your function will be put here*/
int cmp(const void*a,const void*b)
{
    return *(int*)a-*(int*)b;
}
int findHeight(int pre[],int in[],int l,int r,int root)
{
    if(l>r)return 0;
    int cnt=1;
    int i=l;
    while(in[i]!=pre[root])i++;
    int lcnt=findHeight(pre,in,l,i-1,root+1);
    int rcnt=findHeight(pre,in,i+1,r,root+i-l+1);
    if(lcnt<=rcnt)cnt+=rcnt;
    else cnt+=lcnt;
    return cnt;
}
int Height_of_BST(int preorder[],int N)
{
    int pre[N];
    for(int i=0;i<N;i++)
        pre[i]=preorder[i];
    qsort(preorder,N,sizeof(int),cmp);
    return findHeight(pre,preorder,0,N-1,0)-1;
}
```

【考点延伸】 二叉树搜索树